

SE Lab 1 problem statement #32

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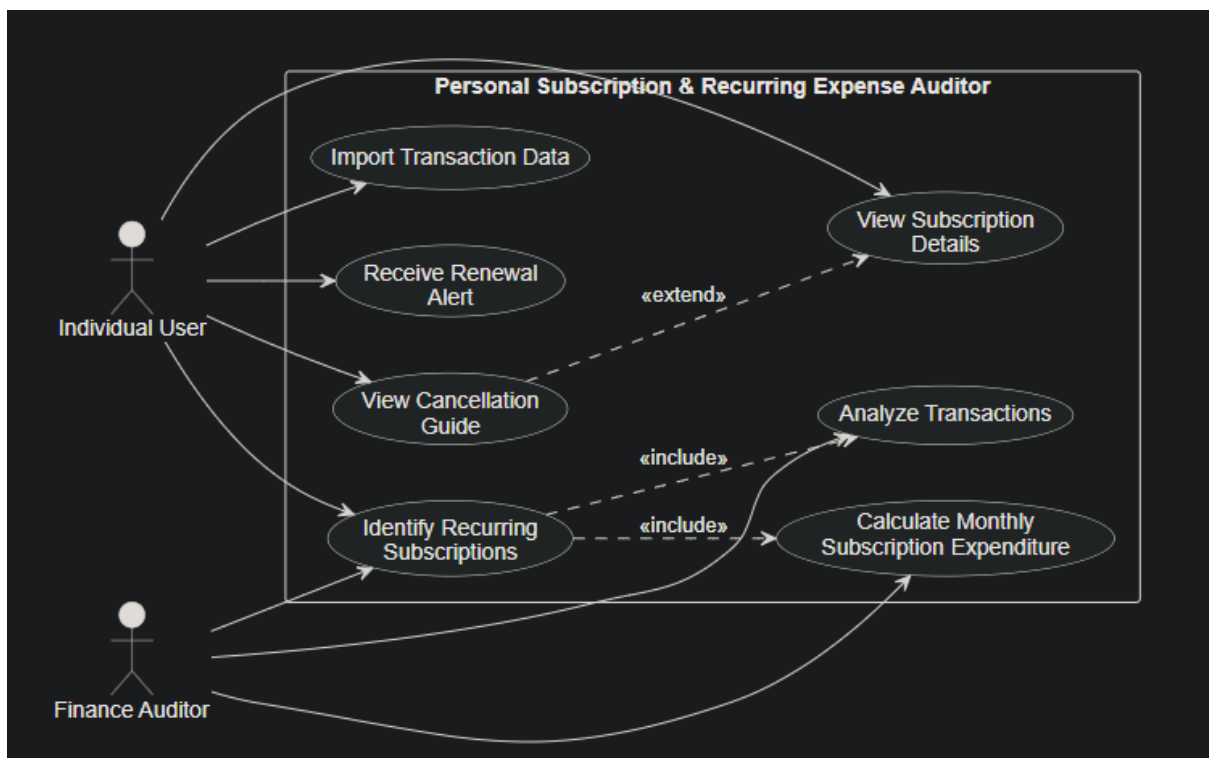
Requirements Table:

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR001	Functional	The system shall analyze imported transactions and identify recurring billing patterns as subscriptions.	High	Recurring service identified and renewal date logged	Identifying recurring expenses is the core purpose of the application.
FR002	Functional	The system shall allow the Individual User to import transaction data from supported formats such as CSV or OFX.	High	Valid transaction file is successfully imported and transactions are displayed. Condition fails when invalid or corrupted file is accepted as valid transaction data.	Transaction data is required for identifying recurring expenses and subscriptions.
FR003	Functional	The system shall calculate and display the user's estimated monthly subscription expenditure based on identified recurring subscriptions.	High	The system calculates the monthly subscription expenditure from identified subscriptions and displays the result. The displayed amount does	Users need to understand their monthly financial commitment to subscriptions.

				not correspond to the identified recurring expenses is a failure.	
FR004	Functional	The system shall identify upcoming subscription renewal dates and alert the Individual User before the renewal date.	Medium	The user receives an alert before an identified subscription renewal. Failure is when no alert is generated for an upcoming renewal.	Advance notification allows users to review subscriptions before being charged again.
FR005	Functional	The system shall provide cancellation guidance for subscriptions selected by the Individual User.	Medium	The system displays cancellation instructions for a selected subscription. Failure is when no cancellation guidance is available for a selected subscription.	Users need an easy way to take action on unwanted recurring subscriptions.
NFR001	Non functional	The system shall process and analyze imported transaction data within an acceptable response time, even when handling a large transaction file.	High	A representative large transaction file is processed within the defined performance target during testing. Failure is when processing time exceeds the defined target under the same test conditions.	Users should not experience excessive delays when importing and analyzing financial transactions.

NFR002	Non functional	The system shall securely process financial transaction imports without storing raw banking credentials in cleartext.	High	Security testing confirms that raw banking credentials are not stored in cleartext and imported financial data is handled securely. Failure is when raw credentials are found stored in cleartext.	Financial information is sensitive, so the system must protect user data and credentials.
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UML diagram:



Use Case Flow Specification:

Use Case ID

UC-003

Use Case Name

Identify Recurring Subscriptions

Primary Actor

Individual User

Supporting Actor

Finance Auditor

Description

This use case describes how the system analyzes imported transaction data and identifies transactions that follow recurring billing patterns, allowing them to be treated as subscriptions.

Preconditions

1. The Individual User has valid transaction data available.
2. The transaction data has been successfully imported into the system.
3. The imported transaction data contains the required transaction information such as date, description, and amount.

Postconditions

1. Recurring transactions are identified and recorded as subscriptions.
2. Each identified subscription has relevant information such as its description, amount, and estimated renewal pattern.
3. The identified subscriptions can be used to calculate the user's recurring monthly expenditure.

Main Success Scenario

1. The Individual User selects the option to identify recurring subscriptions.
2. The system retrieves the imported transaction data.
3. The system analyzes transaction descriptions, amounts, and transaction dates.
4. The system groups transactions that appear to belong to the same recurring service.
5. The system checks the frequency and consistency of the grouped transactions.
6. The system identifies valid recurring billing patterns as subscriptions.
7. The system records the identified subscriptions and their relevant details.
8. The system displays the identified subscriptions to the Individual User.
9. The system makes the identified subscription information available for calculating monthly subscription expenditure.

Alternate Flow – One-Time Payment

1. During transaction analysis, the system encounters a transaction that does not demonstrate a recurring billing pattern.
2. The system classifies the transaction as a one-time payment rather than a subscription.
3. The transaction is excluded from the list of identified recurring subscriptions.

4. The system continues analyzing the remaining transactions.
5. The system displays only transactions that meet the criteria for recurring subscriptions.

Expected Result

The system successfully identifies genuine recurring subscriptions while avoiding incorrect classification of one-time payments.